PLATO: *Privileged Learning from Accurate Trajectory Observations* for IMU Bias Dynamics

Anonymous Author(s)
Affiliation
Address
email

Abstract: Neural inertial odometry has shown strong potential for motion estimation in challenging environments, but most existing methods rely on dense ground-truth trajectories for supervision, which are often unavailable in real robotic deployments. We propose Supervisory Measurement Guided Neural Inertial Odometry, a framework that learns inertial odometry models using sparse but reliable supervisory measurements instead of full trajectory supervision. The key idea is to leverage high-confidence observations, such as loop closures or precise relative pose sources, to guide the learning of neural motion representations. The proposed method integrates neural inertial prediction with a probabilistic filtering framework and formulates training as a bilevel optimization problem that separates trajectory inference from parameter learning. Experiments on simulated and real-world datasets demonstrate that the proposed approach improves odometry accuracy and robustness compared to existing neural inertial odometry baselines.

Keywords: Neural inertial odometry, probabilistic factorization, bilevel optimization

1 Introduction

Inertial measurement units (IMUs) provide high-rate angular velocity and linear acceleration measurements with low cost, low power consumption, and little dependence on external sensing conditions. These properties make inertial odometry attractive for robot state estimation in visually degraded, textureless, smoke-filled, or privacy-sensitive environments. However, inertial-only estimation remains difficult: small gyroscope and accelerometer errors, including time-varying bias and stochastic noise, are repeatedly integrated and quickly lead to severe drift in orientation, velocity, and position.

Recent neural inertial odometry (NIO) methods have shown that learning can significantly improve IMU-only motion estimation. Some methods directly regress motion quantities such as velocity, displacement, or orientation from inertial windows [1, 2, 3, 4, 5, 6], while others retain the inertial motion model and learn IMU corrections, bias, or uncertainty propagation [7, 8, 9, 10]. Despite their success, most existing methods require dense ground-truth trajectories or dense segment-level labels during training. Such labels are convenient in benchmark datasets collected with motion-capture systems or high-grade SLAM, but are often unavailable in real robotic deployments.

In practice, full trajectory supervision may be expensive or impossible to obtain, while sparse but reliable supervisory measurements are often available. Examples include loop closures, occasional relative-pose measurements, fiducial detections, GNSS/RTK fixes, or high-confidence local registration results. These measurements do not label every inertial state, but they contain valuable information about accumulated drift. This paper asks whether such sparse measurements can teach a neural inertial odometry model without requiring dense trajectory ground truth.

Submitted to the 10th Conference on Robot Learning (CoRL 2026). Do not distribute.

Inspired by the privileged learning paradigm in machine learning, we propose *PLATO: Privileged Learning from Accurate Trajectory Observations*, a probabilistic framework that learns IMU bias dynamics for neural inertial odometry by leveraging high-quality supervisory trajectory measurements. The inertial trajectory is treated as a latent variable generated by IMU propagation under learned bias dynamics and learned noise statistics. Supervisory measurements are then incorporated through a marginal likelihood over the latent trajectory. This formulation naturally supports both dense state supervision and sparse loop-closure supervision, while accounting for the uncertainty and cross-time correlations induced by inertial propagation. At test time, the learned model operates from IMU measurements alone.

Our contributions are:

- We formulate NIO training from sparse supervisory measurements as marginal-likelihood maximization over latent inertial trajectories, avoiding the need for dense ground-truth trajectory labels.
- We instantiate the framework with continuous neural IMU bias dynamics and learned noise statistics, allowing supervisory measurements to train both the propagated trajectory mean and covariance.
- We derive an efficient Lie-group invariant-error adjoint for differentiating through inertial propagation and neural bias dynamics on $\mathrm{SE}_2(3)$, enabling end-to-end training with geometrically consistent gradients.

2 Related Work

Learning-based inertial odometry. Learning-based inertial odometry has become a promising alternative to classical strap-down integration. RoNIN introduced a large-scale smartphone inertial navigation benchmark and learned motion estimates from raw IMU sequences under natural device placements [1]. TLIO regresses 3D displacement and covariance, and tightly fuses the learned relative-motion measurements in a stochastic-cloning EKF [2]. IDOL improves pedestrian localization by explicitly learning device orientation before position estimation [3]. RIO introduces rotation-equivariance as an auxiliary supervisory signal and uses test-time training to improve robustness under distribution shift [4]. More recently, AirIO studies UAV inertial odometry with body-frame representations and explicit attitude encoding [5], while Tartan IMU explores foundation-model pretraining and adaptation across diverse robotic platforms [6]. These methods demonstrate the effectiveness of neural motion priors, but their training still largely depends on dense or segment-level motion labels. In contrast, our method learns from sparse supervisory measurements that constrain only selected functions of the latent trajectory.

Neural IMU correction and bias learning. Another line of work combines learning with physical inertial propagation. Brossard et al. learn gyroscope corrections for open-loop attitude estimation [7]. AirIMU learns IMU correction and uncertainty propagation through a differentiable inertial integrator, improving both odometry and downstream pose-graph optimization [8]. Zhou et al. model IMU bias as a conditional distribution using a diffusion model, emphasizing the stochastic nature of bias [9]. Liu et al. model gyroscope and accelerometer bias as continuous neural ODE dynamics and train the model using pose supervision rather than direct bias labels [10]. Our method is closest to continuous neural bias learning, but differs in the supervision regime: instead of assuming dense pose supervision, we train the inertial model from sparse supervisory measurements through a covariance-aware marginal likelihood.

Probabilistic and geometric state estimation. Classical filtering and smoothing methods provide a principled way to combine inertial propagation with external measurements. Invariant filtering exploits the Lie-group structure of robotic dynamics to obtain stable and geometrically consistent estimators [11, 12]. Our gradient derivation builds on this perspective by using invariant errors for IMU propagation on $\mathrm{SE}_2(3)$. We also relate to adjoint-based optimization for neural ODEs [13],

but our setting requires differentiating through manifold-valued inertial dynamics and sparse measurement likelihoods. Unlike conventional state-estimation methods that optimize a trajectory under fixed sensor models, our objective learns the neural inertial model itself from the marginal likelihood of supervisory measurements.

3 Preliminaries

3.1 Matrix Lie Group

Let $\mathbf{G}$ be a matrix Lie group with Lie algebra $\mathfrak{g}$. Note that $\mathfrak{g}$ is isomorphic to $\mathbb{R}^{\dim \mathfrak{g}}$. The exponential map $\exp : \mathbb{R}^{\dim \mathfrak{g}} \to \mathbf{G},\ \xi \mapsto \exp_{\mathrm{m}}(\xi^\wedge)$ relates $\mathfrak{g}$ to its associated group $\mathbf{G}$, where $(\cdot)^\wedge$ denotes the skew operator and $\exp(\cdot) \triangleq \exp_{\mathrm{m}}(\xi^\wedge)$ the matrix exponential. In a neighborhood of the identity, $\exp$ is locally invertible, enabling the definition of the logarithm map $\log : \mathbf{G} \to \mathbb{R}^{\dim \mathfrak{g}},\ X \mapsto \log(X)^\vee$, where $(\cdot)^\vee$ is the inverse of $(\cdot)^\wedge$. The special orthogonal group $SO(3)$ represents the set of all possible rotations of a rigid body in three-dimensional space. The special Euclidean group $SE(3)$ (rigid-body transformations), and $SE_2(3)$ are defined as

$$SE(3) \triangleq \left\{ \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \,\middle|\, R \in SO(3), p \in \mathbb{R}^3 \right\}, SE_2(3) \triangleq \left\{ \begin{bmatrix} R & v & p \\ \mathbf{0} & & I \end{bmatrix} \middle| R \in SO(3), p, v \in \mathbb{R}^3 \right\},$$

where $\mathbf{0}$ and I denote the zero matrix and the identity matrix, with the dimension omitted when clear from context. For $x^\wedge, y^\wedge \in \mathfrak{g}$ and $\|x\|$ small, their compounded exponentials can be approximated to $\exp(x)\exp(y) \approx \exp(\mathrm{dexp}_y^{-1} x + y)$, where dexp_x is the left Jacobian of x. The adjoint operator has the property that $\mathrm{Ad}_{X^{-1}} = (\mathrm{Ad}_X)^{-1}$.

Let $\mathcal{M}$ be the smooth manifold of a Lie group. For any $X_1, X_2 \in \mathcal{M}$ and tangent vector $\xi \in \mathbb{R}^{\dim \mathcal{M}}$, we define the *plus* and *minus* operators, $X_1 \boxplus \xi$ and $X_1 \boxminus X_2$, which map between the manifold and its tangent space. These operators admit left and right versions; the choice is application-dependent and left implicit. For instance, in the left case, $X_1 \boxplus \xi = \exp(\xi) X_1, X_1 \boxminus X_2 = \log(X_1 X_2^{-1})$. The uncertainty of $X \in \mathcal{M}$ is modeled in the tangent space by a Gaussian distribution. Let $\xi \sim \mathcal{N}(0, \Sigma)$, the induced distribution is

$$\mathcal{N}_L(\bar{X}, \Sigma) \triangleq \alpha \exp\left(-\tfrac{1}{2}(X \boxminus \bar{X})^\top \Sigma^{-1} (X \boxminus \bar{X})\right), \tag{1}$$

where α is the normalization coefficient.

3.2 IMU Kinematics

The kinematics of the state is given by

$$\dot{R}_t = R_t\, \omega_t^\wedge\,, \quad \dot{p}_t = v_t, \quad \dot{v}_t = a_t. \tag{2}$$

The increments of the poses are given by IMU, whose measurement model is

$$\tilde{\omega}_t = \omega_t + b_t^g + n_g, \quad \tilde{a}_t = R_t^\top (a_t - g) + b_t^a + n_a, \tag{3}$$

where b_t^g and b_t^a denote the gyroscope and accelerometer bias terms, respectively, n_g and n_a are the corresponding white Gaussian noises, and $\tilde{\cdot}$ denotes the sensor measurements. It is noted that when the state is defined on a Lie group, the above dynamics satisfy a group-affine property.

Determining appropriate values for b_t^a and b_t^g remains a long-standing challenge. Typically, the biases are modeled as a Brownian motion as

$$\dot{b}_t^g = n_{b_g}, \quad \dot{b}_t^a = n_{b_a}, \tag{4}$$

where n_{b_g} and n_{b_a} denote the white noises.

3.3 Invariant Observer

Consider the group-affine systems

$$\dot{X}_t = f_{u_t}(X_t), \tag{5}$$

where u_t is an input variable. Consider two distinct trajectories X_t and $\bar{X}_t$ of (5). The left-invariant and right-invariant errors between the two trajectories are defined by

$$\eta_t = X_t^{-1}\bar{X}_t \ \text{ or } \ \bar{X}_t^{-1}X_t \ \text{ (left invariant)},$$
$$\eta_t = X_t\bar{X}_t^{-1} \ \text{ or } \ \bar{X}_t X_t^{-1} \ \text{ (right invariant)}.$$

According to Theorem 1 in [11], a well-known and useful finding is that the invariant error is state trajectory independent. Let $\xi_t \triangleq \log(\eta_t)$, and it follows the linear differential equation in $\mathbb{R}^d$:

$$\dot{\xi}_t = A_t\xi_t + B_t n_t, \tag{6}$$

where $n_t = [n_t^a; n_t^g]$.

3.4 Continuous Adjoint Method for NODEs

Consider a Neural ODE

$$\dot{x}(t) = f_\theta\big(x(t), u(t)\big), \qquad t \in [t_0, t_N], \tag{7}$$

with initial condition $x(t_0)$. Assume the loss depends on the discrete-time states along the trajectory, i.e., $L = L\big(x(t_0), \ldots, x(t_N)\big)$. To compute $\frac{dL}{d\theta}$ efficiently, the continuous adjoint method introduces the adjoint variable $\alpha(t) \triangleq \frac{\partial L}{\partial x(t)}$ and solves it backward in time. The dynamics of $\alpha(t)$ is

$$\dot{\alpha}(t) = -\left(\frac{\partial f}{\partial x}\big(x(t), u(t); \theta\big)\right)^\top \alpha(t). \tag{8}$$

If the loss contains discrete terms at times $\{t_k\}$, the adjoint satisfies the jump condition $\alpha(t_k^-) = \alpha(t_k^+) + \frac{\partial L}{\partial x(t_k)}$. The gradient with respect to θ is then given by

$$\frac{dL}{d\theta} = \int_{t_0}^{t_N} \alpha(t)^\top \frac{\partial f}{\partial \theta}\big(x(t), u(t); \theta\big)\, dt. \tag{9}$$

4 Problem Formulation

Following Liu et al. [10], we consider $b_t \triangleq \mathrm{vec}(b_t^a, b_t^g)$ to be **deterministic** variables with dynamics modeled by a NODE, as follows:

$$\dot{b}_t = f(b_t, \phi(u_{t-H:t}), t, \theta), \tag{10}$$

where θ denotes the parameters of the NODE to be trained, H denotes the number of historical IMU samples, and $\phi(\cdot)$ is the IMU feature encoder.

Problem 1 *Learn the dynamics of b_t^a and b_t^g in (10) and the covariances of n_a, n_g in (3).*

5 Supervisory Learning Objectives

Given the bias dynamics in (10), the network parameters θ and the IMU noise statistics cannot be learned from direct bias supervision, since the bias trajectory is typically unavailable in practice. Following [10], we instead learn them indirectly from supervisory measurements defined on the propagated motion states.

5.1 Marginal-Likelihood Formulation

Let Y denote the stacked supervisory measurements. We formulate parameter learning as the maximization of the marginal likelihood

$$\hat{\theta} = \arg\max_{\theta \in \Theta} p(Y \mid \theta). \tag{11}$$

To evaluate this likelihood, we introduce the latent trajectory states $X_{1:N}$ generated by lower-level IMU propagation under the learned bias dynamics. Their evolution is modeled as

$$X_{k+1} = f_k(X_k; \theta) + w_k, \qquad w_k \sim \mathcal{N}(\mathbf{0}, Q_k), \tag{12}$$

where Q_k denotes the propagation covariance induced by IMU noise.

Since supervisory measurements are only available at selected time indices, we partition the latent trajectory into two subsets: the supervisory states $\boldsymbol{X}^s$ and the remaining odometry states $\boldsymbol{X}^o$. Then

$$p(Y \mid \theta) = \int p(Y \mid \boldsymbol{X}^s)\, p(\boldsymbol{X}^s \mid \theta)\, \mathrm{d}\boldsymbol{X}^s, \tag{13}$$

where

$$p(\boldsymbol{X}^s \mid \theta) = \int p(X_{1:N} \mid \theta)\, \mathrm{d}\boldsymbol{X}^o. \tag{14}$$

The exact marginalization over $\boldsymbol{X}^o$ is generally intractable. We therefore approximate $p(\boldsymbol{X}^s \mid \theta)$ by a Gaussian distribution obtained from lower-level IMU propagation, with mean $\bar{\boldsymbol{X}}^s$ and covariance $\bar{\boldsymbol{P}}^s$.

5.2 Mean Pretraining and Covariance-Aware Refinement

Let the stacked supervisory residual be

$$\boldsymbol{r} = Y \boxminus h(\bar{\boldsymbol{X}}^s), \qquad \boldsymbol{S} = H \bar{\boldsymbol{P}}^s H^\top + W, \tag{15}$$

where $h(\cdot)$ is the supervisory measurement function, H is the Jacobian of the residual with respect to the supervisory-state error, and W is the supervisory measurement covariance. In training, we use a two-stage objective. We first fit the propagated mean with

$$\theta_{\mathrm{mse}} = \arg\min_\theta L_{\mathrm{MSE}}(\theta), \qquad L_{\mathrm{MSE}}(\theta) = \frac{1}{2} \boldsymbol{r}^\top \boldsymbol{r}, \tag{16}$$

with covariance parameters fixed or initialized to nominal values. Starting from θ_{mse}, we optimize the covariance-aware likelihood

$$L_{\mathrm{NLL}}(\theta) = \frac{1}{2} \boldsymbol{r}^\top \boldsymbol{S}^{-1} \boldsymbol{r} + \frac{1}{2} \log\det \boldsymbol{S}, \tag{17}$$

jointly over the bias dynamics and noise statistics. The remaining question is how to evaluate $\boldsymbol{S}^{-1}$ and $\log\det \boldsymbol{S}$ without forming the dense marginal covariance. We instantiate this next for pose-trajectory supervision.

5.3 Trajectory State Supervision

Consider a high-accuracy supervisory trajectory available at timestamps $t_i^s = t_{\kappa_i}$, $i = 0, \ldots, M$. The lower level remains a pure IMU invariant preintegration under the learned bias dynamics; the supervisory trajectory only specifies the measurement function in (17). Let $\bar{X}_i^s \in SE_2(3)$ be the propagated mean and write the right-invariant perturbation as

$$X_i^s = \exp((\xi_i^s)^\wedge)\, \bar{X}_i^s, \qquad \boldsymbol{\xi}^s = [(\xi_0^s)^\top, \ldots, (\xi_M^s)^\top]^\top. \tag{18}$$

For IMU propagation on $SE_2(3)$, the right-invariant error is log-linear:

$$\dot{\xi}_t = F \xi_t + G_t n_t, \qquad F = \begin{bmatrix} 0 & 0 & 0 \\ g^\wedge & 0 & 0 \\ 0 & I & 0 \end{bmatrix}. \tag{19}$$

Thus $\mathcal{F}$ is trajectory independent and time invariant, with analytic transition

$$\Phi(\Delta t) = \exp(\mathcal{F} \Delta t) = \begin{bmatrix} I & 0 & 0 \\ g^\wedge \Delta t & I & 0 \\ \frac{1}{2} g^\wedge \Delta t^2 & \Delta t I & I \end{bmatrix}. \tag{20}$$

Marginalizing the IMU-rate states between supervised timestamps gives the downsampled Gaussian chain

$$\xi^s_{i+1} = \Phi^s_i \xi^s_i + \nu^s_i, \qquad \nu^s_i \sim \mathcal{N}(0, Q^s_i), \qquad \Phi^s_i = \Phi(t^s_{i+1} - t^s_i), \tag{21}$$

where Q^s_i is the preintegrated covariance over $[t^s_i, t^s_{i+1}]$ computed with (20) and $\mathcal{G}_t$. The covariance $\bar{\boldsymbol{P}}^s$ of $\boldsymbol{\xi}^s$ is dense because all supervised states share the same inertial integration history. However, the precision $\bar{\boldsymbol{\Lambda}}^s = (\bar{\boldsymbol{P}}^s)^{-1}$ is block tridiagonal. Its nonzero blocks are

$$\begin{aligned}
\bar{\Lambda}^s_{0,0} &= (P^s_0)^{-1} + (\Phi^s_0)^\top (Q^s_0)^{-1} \Phi^s_0, \\
\bar{\Lambda}^s_{i,i} &= (Q^s_{i-1})^{-1} + (\Phi^s_i)^\top (Q^s_i)^{-1} \Phi^s_i, && i = 1, \dots, M-1, \\
\bar{\Lambda}^s_{M,M} &= (Q^s_{M-1})^{-1}, \\
\bar{\Lambda}^s_{i,i+1} &= -(\Phi^s_i)^\top (Q^s_i)^{-1}, \qquad \bar{\Lambda}^s_{i+1,i} = -(Q^s_i)^{-1} \Phi^s_i.
\end{aligned} \tag{22}$$

This precision is the object used in training; the dense covariance $\bar{\boldsymbol{P}}^s$ is never explicitly formed.

Pose-trajectory likelihood. Let $Y^p_i \in SE(3)$ be the supervisory pose at t^s_i, and let $\pi_p(X^s_i) \in SE(3)$ extract the pose from X^s_i. We define

$$r^p_i = \log\!\left(Y^p_i \, \pi_p(\bar{X}^s_i)^{-1}\right), \qquad \boldsymbol{r}^p = [(r^p_0)^\top, \dots, (r^p_M)^\top]^\top. \tag{23}$$

Linearizing the pose residual gives

$$\boldsymbol{r}^p = H^p \boldsymbol{\xi}^s + e^p, \qquad e^p \sim \mathcal{N}(0, W^p), \tag{24}$$

where H^p is block diagonal and $W^p = \mathrm{blkdiag}(W^p_0, \dots, W^p_M)$. For nearly noiseless pose supervision, W^p is kept small but positive definite. Therefore

$$\boldsymbol{r}^p \sim \mathcal{N}(0, \boldsymbol{S}^p), \qquad \boldsymbol{S}^p = H^p \bar{\boldsymbol{P}}^s (H^p)^\top + W^p, \tag{25}$$

and the pose-trajectory NLL is

$$L_{\mathrm{pose}}(\theta) = \frac{1}{2} (\boldsymbol{r}^p)^\top (\boldsymbol{S}^p)^{-1} \boldsymbol{r}^p + \frac{1}{2} \log\det \boldsymbol{S}^p. \tag{26}$$

Using Woodbury and the determinant lemma, define

$$K^p = \bar{\boldsymbol{\Lambda}}^s + (H^p)^\top (W^p)^{-1} H^p, \qquad \eta^p = (H^p)^\top (W^p)^{-1} \boldsymbol{r}^p. \tag{27}$$

Then (26) is evaluated as

$$\begin{aligned}
L_{\mathrm{pose}}(\theta) = & \frac{1}{2} (\boldsymbol{r}^p)^\top (W^p)^{-1} \boldsymbol{r}^p - \frac{1}{2} (\eta^p)^\top (K^p)^{-1} \eta^p \\
& + \frac{1}{2} \log\det W^p + \frac{1}{2} \log\det K^p - \frac{1}{2} \log\det \bar{\boldsymbol{\Lambda}}^s.
\end{aligned} \tag{28}$$

Full state supervision is recovered by replacing π_p with the identity map and H^p with the identity matrix.

Under the two-stage schedule above, the MSE stage uses $L^p_{\mathrm{MSE}} = \frac{1}{2} (\boldsymbol{r}^p)^\top \boldsymbol{r}^p$ and the NLL stage uses (28). Compared with MSE, NLL adds covariance propagation over the same IMU intervals and sparse factorizations for $\bar{\boldsymbol{\Lambda}}^s$ and K^p. With fixed state dimension $d_x = \dim SE_2(3)$, the additional cost is $O(N d_x^3 + M d_x^3)$ time and $O(M d_x^2)$ storage, where N is the number of IMU samples. The analytic transition in (20) avoids numerical matrix exponentials, and the block-tridiagonal precision avoids the dense $O((M d_x)^3)$ likelihood. Since d_x is small and NLL is used only after MSE pretraining, this refinement cost is acceptable.

6 Gradient Estimation

By the chain rule, the effect of θ flows from $b_k = \mathrm{vec}(b_a, b_g)$ through T_k to loss $\mathcal{L}$. The gradient computation follows several steps. First, consider the estimated trajectory error as an **implicit function of** b_k, then compute the gradient $\frac{\partial \mathcal{L}}{\partial b_k}$ through backpropagation as discussed in 6.1. Second, using the adjoint method to compute $\frac{\partial \mathcal{L}}{\partial \theta}$ with detail in 6.2.

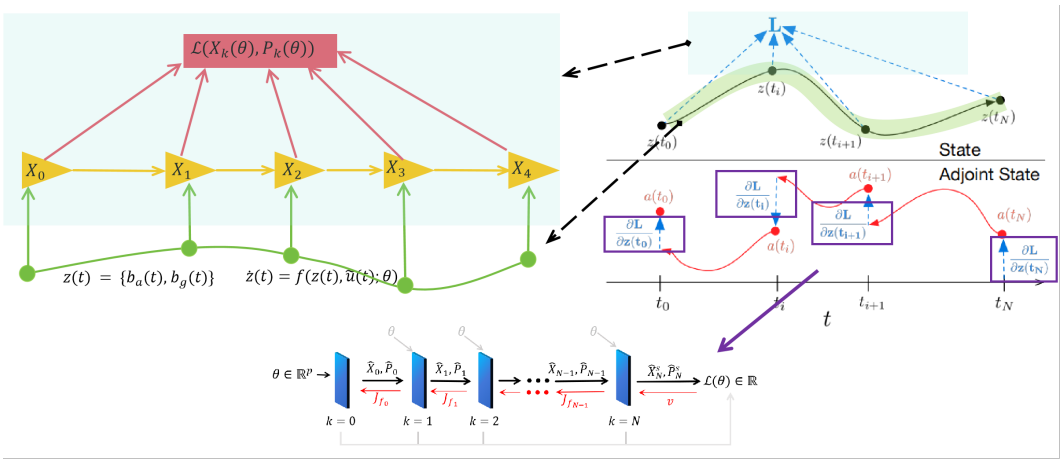


Figure 1: Modules of the algorithm.

6.1 Outer Adjoint for Bias Sensitivity

We analyze the sensitivity of the invariant error dynamics with respect to the IMU bias. Let $\xi_k \in \mathbb{R}^d$ denote the right-invariant error at time step k. Recall that $n_t = [n_t^a; n_t^g]$. Under first-order linearization, the error propagation admits the form

$$\xi_{k+1} = F_k \xi_k + G_k n_k,$$

where F_k and G_k are the state transition and noise Jacobians. By adopting the right-invariant error, the continuous-time dynamics in (6) become state-trajectory independent, and A_t is constant:

$$A_t = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ g^\wedge & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I & \mathbf{0} \end{bmatrix}.$$

According to the linearization of the IMU kinematics, G_k and F_k take the form

$$G_k = -\Delta t_k \begin{bmatrix} 0 & \bar{R}_k \\ \bar{R}_k & (\bar{v}_k)^\wedge \bar{R}_k \\ 0 & (\bar{p}_k)^\wedge \bar{R}_k \end{bmatrix}, \qquad F_k = \exp(A_t \Delta t_k) = \begin{bmatrix} I & 0 & 0 \\ g^\wedge \Delta t_k & I & 0 \\ \frac{1}{2} g^\wedge \Delta t_k^2 & I \Delta t_k & I \end{bmatrix}.$$

To optimize the bias, the error system of interest is

$$\xi_{k+1} = F_k \xi_k + M_k \delta b_k. \tag{29}$$

Since the noise term depends explicitly on the bias $b_k = \mathrm{vec}(b_k^a, b_k^g)$, and we maintain the ordering

$$n_k = \begin{bmatrix} \tilde{a}_k - R_k^\top (a_k - g) - b_k^a \\ \tilde{\omega}_k - \omega_k - b_k^g \end{bmatrix}, \qquad \frac{\partial n_k}{\partial b_k} = -I,$$

it follows that

$$M_k = G_k \frac{\partial n_k}{\partial b_k} = -G_k.$$

Note that ξ_k depends only on the past biases $b_{0:k-1}$ and is independent of b_k.

We consider a stage-wise loss defined on the invariant error:

$$L = \sum_{k=1}^{N} \ell_k(\xi_k). \tag{30}$$

Define the adjoint variable as $\lambda_k \triangleq \frac{\partial L}{\partial \xi_k}$, $\lambda_N = \frac{\partial \ell_N}{\partial \xi_N}$. Applying the chain rule to (29), the adjoint satisfies the backward recursion

$$\lambda_k = \frac{\partial \ell_k}{\partial \xi_k} + F_k^\top \lambda_{k+1}. \tag{31}$$

Finally, the gradient of the total loss with respect to the bias perturbation is

$$\frac{\partial L}{\partial \delta b_k} = \frac{\partial L}{\partial \xi_{k+1}} \frac{\partial \xi_{k+1}}{\partial \delta b_k} = M_k^\top \lambda_{k+1} = -G_k^\top \lambda_{k+1}. \tag{32}$$

Remark. While neural ODE frameworks are typically developed in Euclidean spaces and rely on automatic differentiation through nonlinear dynamics, directly applying them to Lie group systems is nontrivial due to the manifold structure. Instead of differentiating through the nonlinear IMU kinematics on $\mathrm{SE}_2(3)$, we leverage the group-affine property of the system. By introducing the invariant error $\xi = \log(\eta)$, the error dynamics admit a log-linear representation. This allows us to construct a linear surrogate system in the Lie algebra, through which gradients can be propagated efficiently and in closed form.

6.2 Inner Adjoint Method for NODE Training

We have modeled the bias dynamics as NODE with network parameter θ. To optimize θ, the b_k can be considered as the latent variables. After computing the $\frac{\partial L}{\partial b_k}$ in the subsection 6.1, we can use adjoint method to compute the gradient $\frac{\partial L}{\partial \theta}$ as described in 3.4.

7 Experimental

7.1 Pre-training Diagnostic

PLATO is not tied to a manually fixed IMU input representation. We provide a pre-training diagnostic protocol to select an IMU feature encoder according to the sensor characteristics, motion spectrum, and validation rollout behavior. In this subsection, we will present the IMU feature encoder for the two distinct dataset used in the experient.

7.2 Euroc Dataset

The integration path should be relatively short. The motion pattern should have diversity.

The double adjoint methods exceed in memory.

7.3 Application in the Underwater VIO

Dataset setting:

8 Dicussion and Limitation

9 Conclusion

Table 1: Evaluation comparison on EuRoC test sequences. Lower is better for all metrics.

Method	Done	AOE ↓	ATE Vel. ↓	ATE Pos. ↓
Euroc_SE23_w64	5/5	3.494	**16.070**	**672.284**
Euroc_baseline_w64	5/5	3.494	16.295	678.010
Euroc_SE23decay_w64	5/5	3.494	16.214	678.145
Euroc_SE23_NLL_optQ_gt_random_w64	5/5	3.392	16.319	693.231
Euroc_SE23_w128	5/5	**2.552**	18.239	735.463
Euroc_SE23_w96	5/5	2.714	18.206	738.347
Euroc_baseline_w96	5/5	2.714	18.344	743.180
Euroc_baseline_w128	5/5	**2.552**	18.222	744.752
Euroc_piecewise_gt_bias_5hz	5/5	2.979	18.878	757.356
Euroc_SE23decay_lam1p5_w96	5/5	2.714	18.799	763.051
Euroc_SE23_NLL_optQ_gt_random_w16	5/5	2.714	17.818	768.651
Euroc_baseline_w16	5/5	2.714	17.871	771.703
Euroc_SE23_w16	5/5	2.714	18.068	779.317
Euroc_SE23_NLL_fixedcov_gt_random_w16	5/5	2.714	18.188	789.343
Euroc_SE23decay_w16	5/5	2.714	18.245	790.616
Euroc_SE23_vo_body_w64_noV203	5/5	3.461	21.077	813.245
Euroc_SE23_vo_body_source_dense_balanced_w8_noV203	5/5	4.513	18.155	815.423
Euroc_SE23_w32	5/5	2.671	20.492	826.662
Euroc_baseline_w32	5/5	2.671	20.636	833.129
Euroc_SE23_vo_body_source_dense_w4_noV203	5/5	4.926	18.752	841.925
Euroc_SE23decay_w32	5/5	2.671	20.837	842.971
Euroc_SE23_vo_body_w32_noV203	5/5	4.863	21.270	889.392
Euroc_SE23_vo_body_source_dense_w8_noV203	5/5	3.527	20.747	893.384
Euroc_SE23_vo_body_source_dense_mix_w8_noV203	5/5	3.527	20.757	899.625
Euroc_SE23_vo_body_source_w4_noV203	5/5	5.625	26.082	987.707
Euroc_SE23_vo_body_source_w16_noV203	5/5	5.767	27.880	1091.974
Euroc_SE23_vo_body_w16_noV203	5/5	4.776	27.792	1186.027
Euroc_piecewise_gt_bias_1hz	5/5	5.882	46.103	1958.631
Euroc_constant_first_gt_bias	5/5	23.219	142.666	5972.669

Table 2: Training time and GPU memory comparison for window size 64.

Method	Train Time (s)	Avg. Epoch Time (s)	Max GPU Mem. (MB)	Acc. Avg. Epoch (s)
Baseline AD	**2606.77**	**0.723**	1214.59	**0.697**
Analytic Adjoint	2674.06	0.742	**403.63**	0.780

References

[1] S. Herath, H. Yan, and Y. Furukawa. Ronin: Robust neural inertial navigation in the wild: Benchmark, evaluations, & new methods. In *2020 IEEE international conference on robotics and automation (ICRA)*, pages 3146–3152. IEEE, 2020.

[2] W. Liu, D. Caruso, E. Ilg, J. Dong, A. I. Mourikis, K. Daniilidis, V. Kumar, and J. Engel. Tlio: Tight learned inertial odometry. *IEEE Robotics and Automation Letters*, 5(4):5653–5660, 2020.

[3] S. Sun, D. Melamed, and K. Kitani. Idol: Inertial deep orientation-estimation and localization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 6128–6137, 2021.

[4] X. Cao, C. Zhou, D. Zeng, and Y. Wang. Rio: Rotation-equivariance supervised learning of robust inertial odometry. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6614–6623, 2022.

[5] Y. Qiu, C. Xu, Y. Chen, S. Zhao, J. Geng, and S. Scherer. Airio: Learning inertial odometry with enhanced imu feature observability. *IEEE Robotics and Automation Letters*, 2025.

[6] S. Zhao, S. Zhou, R. Blanchard, Y. Qiu, W. Wang, and S. Scherer. Tartan imu: A light foundation model for inertial positioning in robotics. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 22520–22529, 2025.

[7] M. Brossard, S. Bonnabel, and A. Barrau. Denoising imu gyroscopes with deep learning for open-loop attitude estimation. *IEEE Robotics and Automation Letters*, 5(3):4796–4803, 2020.

[8] Y. Qiu, C. Wang, C. Xu, Y. Chen, X. Zhou, Y. Xia, and S. Scherer. Airimu: Learning uncertainty propagation for inertial odometry. *arXiv preprint arXiv:2310.04874*, 2023.

[9] S. Zhou, S. Katragadda, and G. Huang. Learning imu bias with diffusion model. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2162–2168. IEEE, 2025.

[10] B. Liu, T.-Y. Lin, W. Zhang, and M. Ghaffari. Debiasing 6-dof imu via hierarchical learning of continuous bias dynamics. In *Robotics: Science and Systems (RSS)*, 2025.

[11] A. Barrau and S. Bonnabel. The invariant extended kalman filter as a stable observer. *IEEE Transactions on Automatic Control*, 62(4):1797–1812, 2016.

[12] R. Hartley, M. Ghaffari, R. M. Eustice, and J. W. Grizzle. Contact-aided invariant extended kalman filtering for robot state estimation. *The International Journal of Robotics Research*, 39 (4):402–430, 2020.

[13] J. Zhuang, N. Dvornek, X. Li, S. Tatikonda, X. Papademetris, and J. Duncan. Adaptive checkpoint adjoint method for gradient estimation in neural ode. In *International Conference on Machine Learning*, pages 11639–11649. PMLR, 2020.